Adaptive Deep Learning for Physics Discovery: Enabling Rapid Experiment-Specific Scinet Search

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INTRODUCTION

Throughout history, physicists have condensed complex observations into abstract physical laws allowing predictions and theoretical understanding such as Newton laws. Recent advancements, like the SciNet neural network architecture, have revolutionized physics by emulating the physicist's reasoning process, making predictions and rediscovering fundamental laws solely based on observations and without prior knowledge in physics.
In this poster, we introduce a novel methodology to enhance SciNet and adapt it to a wide range of physical experiments.

MOTIVATION

Making machine learning in the physical reasoning process more accessible to scientists, plus enhancing its use including the SciNet's performance improvement.

OBJECTIVE

Develop a general, experiment-specific and automated machine learning framework to enhance the physical reasoning process for physicians and scientists.

SCINET

SciNet is a neural network architecture used in the physical reasoning and discovery process without relying on prior physical assumptions.
Specifically, it extracts relevant latent variables to answer physical questions using only experimental data. One of the most significant examples is the ability of a SciNet-based machine learning model to discover our solar system heliocentricity [1].

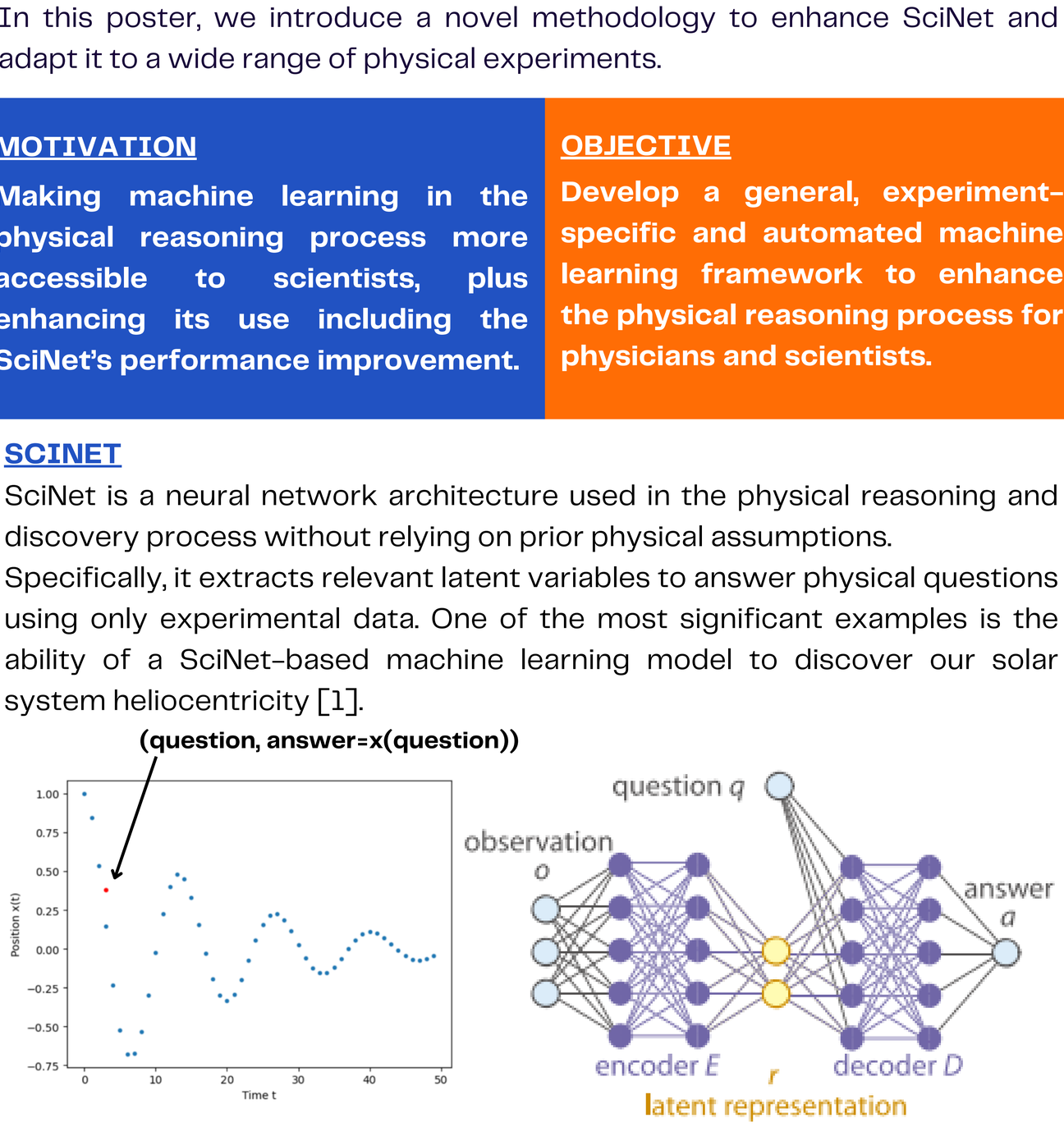


FIGURE 1: SciNet architecture

METHODOLOGY

Our methodology involves SciNet generalization through hyperparameter tuning. This tuning uses random search on a space of relevant hyperparameters to optimize the model, make predictions and minimize the similarity between the extracted latent variables.

Hyperparameter type	Hyperparameter	Description	Range
Architecture	# layers	Number of layers in each of the encoder and decoder	[2, 3, 4, 5, 6]
	# neurons	Number of neurons in each layer of the encoder/decoder	[50, 100, 200, 250, 300, 500]
	Type of activation	Type of activation applied after each layer	[Relu, Gelu, Relu6, leakyRelu]
Experiment-specific	Size	The time series input size	[5,10,25,50, 100, 150]
	# latent neurons	Number of relevant variables ent representation	[1, 2, 3, 4]

TABLE I: Summary of the search space design. # stands for "number of"

- **Search Space:** The search space comprises two types of hyperparameters : architecture and experiment-specific. The experiment-specific are highly dependent on the explored physical phenomenon. Table I summarizes the search space.
- **Objective:** A key behavior of Scinet is to build a latent representation that represents physical factors resulting in the observation. To achieve this, the latent representation neurons have to be independent. This objective is coupled with the original prediction performance computed with the mean absolute error.

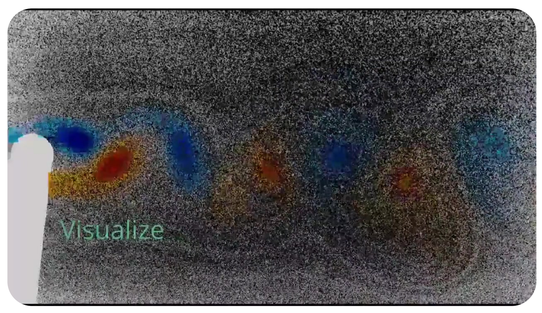
FLUID PARTICLE MOTION

In assessing our methodology, we focused on the particle motion within a non-uniform fluid. We generated multiple time-series as a dataset using equation [3], and subsequently fine-tuned and trained SciNet using our hyperparameter optimization method.

$$f(t, st, fr) = st^2 \cdot fr \cdot (1 - \frac{t}{st} - exp(-\frac{t}{st}))$$

t : Time, st : Drag Force
fr : Froude Number (Gravitational Force)

FIGURE 3: Motion equation



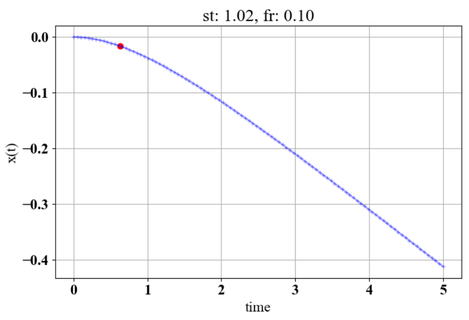


FIGURE 2: Generated experimental time-serie dataset.

The analysis of the latent representation in Figure [4] indicates that the model successfully identified relevant variables, as evidenced by the correlation between the neurons' output and pertinent variables.

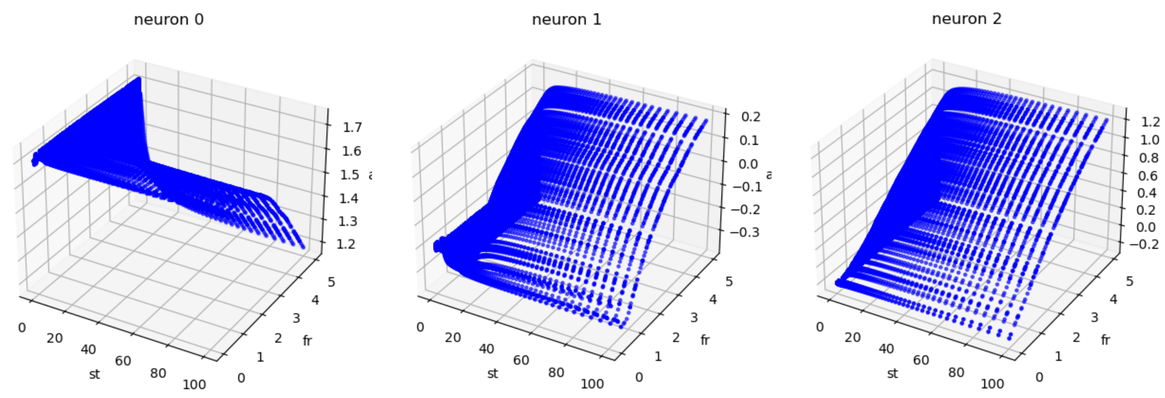


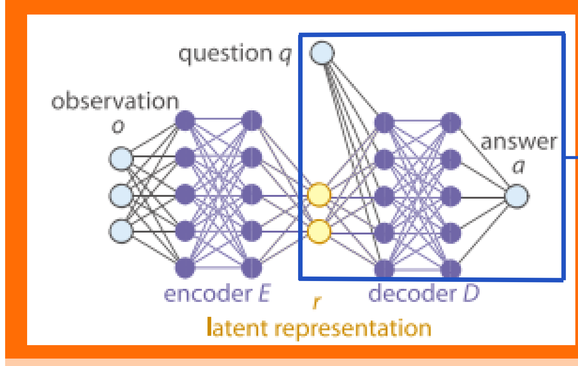
FIGURE 4: Relation between the latent representation and relevant variables

CONCLUSION

We proposed an experiment-based hyperparameter optimization procedure for SciNet to generelize it to any physical experiment and improve its performance in extracting relevant variables for addressing our physical questions.

FUTURE WORKS

As a second step towards achieving our goal, we aim to integrate a mathematical formulation component into the framework. This will empower scientists not only to obtain relevant variables but also to derive a simpler algebraic formula describing the relationships between these variables., accomplished through the interpretation of our decoder neural network inspired by [2].



$$F = \frac{\alpha x_1 x_2}{{x_3}^2} \approx \frac{G M_1 M_2}{r^2}$$

REFERENCES

[1] Iten, Raban, et al. "Discovering physical concepts with neural networks." Physical review letters 124.1 (2020): 010508.
[2] Lample, G., & Charton, F. (2019). Deep learning for symbolic mathematics. arXiv preprint arXiv:1912.01412.

This work was done during School of AI Algiers Research summer camp 2023